

This sprint outlines the tasks required to complete project sprint. The dataset will be updated, the Gaussian HMM will be limited to statistically appropriate features (R1, R2, log-transformed V1), the DTW Hierarchical clustering will be computed, and all the conclusions and comparisons will be drawn from the results and written onto the paper. Each task includes explicit deliverables. For any doubts, refer to the discord.

(1) Data Acquisition and Feature Engineering

- (a) Retrieve 3 years of daily (approx. 800 days) SP500 data (adjusted close). Verify completeness and consistency of timestamps.
- (b) Clean the dataset rigorously.
- (c) Construct the following Gaussian-compatible features:
 - **R1:** daily log returns, $\ln(P_t) - \ln(P_{t-1})$
 - **R2:** 5-day log returns, $\ln(P_t) - \ln(P_{t-5})$
 - **V1:** 14-day realized volatility (annualized), log-transformed $V1' = \log(V1)$
- (d) Standardize all features (mean 0, standard deviation 1). Verify distributions visually and statistically.
- (e) Implement Gaussian HMM:
 - Fit HMM with 2–4 states.
 - Extract hidden state sequence, state means, and variances.
 - Validate that the model captures coherent volatility regimes.

(2) Regime Detection via Hierarchical DTW Clustering

- (a) Segment each feature into rolling windows of 21 trading days. This assumes (to be explicitly documented in the paper) that typical regimes span approximately this duration.
- (b) Construct a multivariate array of shape $(n_windows, window_length, n_features)$, where each window contains all selected Gaussian features.

Tip: Use the `clustering_class.py` submodule from `helpermodules` in BSQ's github page. It provides pre-built functions for windowing, multivariate DTW distance computation, and hierarchical clustering.

- (c) Compute the multivariate DTW distance matrix between all windows.
- (d) Apply hierarchical clustering. Plot dendrograms and cut the tree to define 2–4 clusters, each representing a distinct regime.
- (e) Map cluster labels back to the original timeline and overlay them on SP500 price charts. Validate visually and verify consistency with known market behavior.

(3) Financial & Statistical Analysis and Interpretation of HMM vs Clustering Results

- (a) Compare HMM state sequences with DTW clusters:

- Overlay regimes on SP500 price charts.
 - Compute feature statistics per regime (mean, volatility, drawdown).
- (b) Provide financial interpretation:
- Bull regime: positive returns, low volatility.
 - Bear regime: negative returns, high volatility.
 - Confirm if clustering and HMM regimes align.
- (4) Cross-Feature Regime Validation**
- (a) Compare identified regimes with additional dataset features (cross-asset, curve, credit indicators).
- (b) Confirm internal consistency:
- Bull regime: all related features coherent with upward trends.
 - Bear regime: all related features coherent with risk-off behavior.
- (c) Document discrepancies and highlight potential causes.
- (5) Notebook Consolidation and Execution**
- (a) Produce two notebooks:
- HMM analysis notebook: calls pre-existing functions, fits HMM, plots regimes.
 - DTW clustering notebook: calls pre-existing functions, computes DTW distances, performs hierarchical clustering, plots regimes.
- (b) Include clear markdown explanations for each step, describing intent and methodology.
- (c) Ensure repository organization.
- (6) Overleaf Report Development**
- (a) Update or write the following sections:
- Introduction: articulate the motivation, context, and problem statement.
 - Abstract: concise summary of methods, key results, and financial implications.
 - Model pipeline: diagram and explanation of full workflow (data $\rightarrow$ features $\rightarrow$ HMM $\rightarrow$ clustering $\rightarrow$ interpretation).
 - Data collection pipeline: updated detailed description of data retrieval and feature computation.
- (b) Ensure all figures are correctly imported and rendered.
- (c) Provide interpretation and discussion of HMM vs clustering, emphasizing insights relevant to volatility regimes.